

BOSTON TEA PARTY 2.0

‘A new way to file your taxes’ · Boston, Massachusetts

1. Overview · 2. Data · 3. Architecture · 4. Evaluation

I -- THE PROBLEM & PURPOSE

MILLIONS OF TAXPAYERS NAVIGATE A COMPLEX SYSTEM WITH LITTLE TO NO GUIDANCE

We are building a constraint driven and source grounded AI companion that helps everyday filter tax rules, no guesswork when filing taxes

THE PROBLEM

The U.S. tax code is a deeply layered system governed by statute, administrative guidance, and court rulings. For most people, figuring out which rules apply to their situation, their filing status, deductions, work across multiple states is genuinely difficult.

Small changes in circumstance can completely change what deductions you qualify for, what documents you need, and how you should file. These relationships are rarely obvious to non-experts.

“No taxation without comprehension and understanding”

Our system Boston Tea Party 2.0 is a constraint-aware, agentic tax filing companion that helps users understand tax rules and prepare for filing, without replacing a licensed professional.

DESIGN GOALS

- Educational, Not Advisory**
Explains rules clearly. Does not file or give legal advice.
- Source-Grounded**
Every claim is tied to a real, citable authority.
- Constraint-Driven**
Hard eligibility rules are enforced deterministically not guessed by the LLM.
- Dynamic Replanning**
Updates coherently as the user's fact pattern changes.

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II -- THE DATA

THREE AUTHORITATIVE SOURCES POWER THE KNOWLEDGE BASE

Our RAG pipeline draws from statute, administrative guidance, and the courts, the same sources a tax professional would consult.

SOURCE I	SOURCE II	SOURCE III
Internal Revenue Code <i>Statutory Law</i> The primary legal foundation of federal taxation. The IRC defines eligibility rules, deduction limits, and filing requirements. It is precise, structured, and hierarchical, organized by title, chapter, subchapter, and section. - Primary legal authority - Hierarchical §§ structure - Hard eligibility rules	IRS Publications <i>Administrative Guidance</i> The IRS translates statute into taxpayer-facing language through publications like Pub. 17 and Pub. 501. These include plain-language explanations, worked examples, and are released in tax-year-specific editions. - Plain-language summaries - Tax-year versioned - Worked examples & edge cases	U.S. Tax Court Opinions <i>Judicial Interpretation</i> When statute and IRS guidance leave ambiguity, courts step in. Tax Court opinions resolve real contested scenarios and clarify where the law's edges actually fall, essential for handling non-standard fact patterns. - Judicial interpretation - Precedent-setting rulings - Contested fact patterns
<i>All retrieval is grounded in authoritative sources. No facts fabricated. No statutes invented. Citations required.</i>		

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III -- THE ARCHITECTURE

DETERMINISTIC CONSTRAINT REASONING MEETS RETRIEVAL-AUGMENTED GENERATION

The constraint engine decides eligibility. The language model explains it. These responsibilities are deliberately kept separate.



WHY SEPARATE CONSTRAINT LOGIC FROM THE LLM?

LLMs are good at explaining but they are not reliable for enforcing hard legal rules. By running eligibility decisions through a **deterministic constraint engine**, we guarantee that legal boundaries are never softened by probabilistic generation.

REPLANNING ON FACT CHANGES

When a user updates their situation like a new state of employment, a change in deduction strategy, the planning agent **re-evaluates all constraints from scratch**, keeping eligibility assessments and documentation requirements consistent throughout the session.

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IV -- THE EVALUATION

THREE QUESTIONS: IS IT CORRECT, IS IT GROUNDED, DOES IT ACTUALLY HELP?

We evaluate across constraint accuracy, retrieval faithfulness, and real user comprehension, benchmarked against a vanilla LLM baseline.

I. CONSTRAINT ACCURACY

- Eligibility classification accuracy vs. expert-labeled ground truth
- False positive / negative rates on filing path decisions
- Coverage of edge cases: multi-state work, varied filing status

II. RAG FAITHFULNESS

- Citation grounding rate: what % of claims link to a retrieved source
- Hallucination detection via source-answer consistency checks
- Tax-year accuracy: is the correct publication edition being retrieved?

III. USER COMPREHENSION

- Task completion rate on realistic taxpayer scenarios
- Pre/post comprehension scoring
- Perceived trustworthiness and helpfulness ratings

BENCHMARK DESIGN

A curated set of **realistic taxpayer fact patterns** like charitable deductions, multi-state work, itemized vs. standard deductions each with expert-verified correct answers.

BASELINE COMPARISON

All results compared against a vanilla LLM (no constraints, no RAG) to show the concrete gains from our hybrid architecture.

"Success: hard rules enforced correctly, every claim backed by a source, and users who understand more than when they started."